Patients with high-grade gliomas harboring deletions of chromosomes 9p and 10q benefit from temozolomide treatment.

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Surgical cure of glioblastomas is virtually impossible and their clinical course is mainly determined by the biologic behavior of the tumor cells and their response to radiation and chemotherapy. We investigated whether response to temozolomide (TMZ) chemotherapy differs in subsets of malignant glioblastomas defined by genetic lesions. Eighty patients with newly diagnosed glioblastoma were analyzed with comparative genomic hybridization and loss of heterozygosity. All patients underwent radical resection. Fifty patients received TMZ after radiotherapy (TMZ group) and 30 patients received radiotherapy alone (RT group). The most common aberrations detected were gains of parts of chromosome 7 and losses of 10q, 9p, or 13q. The spectrum of genetic aberrations did not differ between the TMZ and RT groups. Patients treated with TMZ showed significantly better survival than patients treated with radiotherapy alone (19.5 vs 9.3 months). Genomic deletions on chromosomes 9 and 10 are typical for glioblastoma and associated with poor prognosis. However, patients with these aberrations benefited significantly from TMZ in univariate analysis. In multivariate analysis, this effect was pronounced for 9p deletion and for elderly patients with 10q deletions, respectively. This study demonstrates that molecular genetic and cytogenetic analyses potentially predict responses to chemotherapy in patients with newly diagnosed glioblastomas.

Functional and molecular analyses of 10q deletions in human gliomas.

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Extensive genomic deletions involving chromosome 10 are the most common genetic alteration in glioblastoma multiforme (GBM). To localize and examine the potential roles of two chromosome arm 10q tumor suppressor regions, we used two independent strategies: mapping of allelic deletions, and functional analysis of phenotypic suppression after transfer of chromosome 10 fragments. By allelic deletion analysis, the region of 10q surrounding the MMAC/PTEN locus was shown to be frequently lost in GBMs but maintained in most low-grade astrocytic tumors. An additional region at 10q25 containing the DMBT1 locus was lost in all grades of gliomas examined. The potential biological significance of these two regions was further assessed by examining microcell hybrids that contained various fragments of 10q. Somatic cell hybrid clones that retained the MMAC/PTEN locus have a less transformed phenotype with clones exhibiting an inability to grow in soft agarose. However, presence or absence of DMBT1 did not correlate with any in vitro phenotype assessed in our model system. These results support a model of molecular progression in gliomas in which the frequent deletion of 10q25-26 is an early event and is followed by the deletion of the MMAC/PTEN during the progression to high-grade GBMs.

Distinct patterns of deletion on 10p and 10q suggest involvement of multiple tumor suppressor genes in the development of astrocytic gliomas of different malignancy grades.

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Deletions of chromosome 10 are the most frequent genetic abnormality in glioblastomas. Several commonly deleted regions have been proposed; however, they are not coincident. We have deletion mapped chromosome 10 in 198 astrocytic gliomas using 53 microsatellite markers. Two commonly deleted regions on 10p were identified, one of which lies between D10S594 and D10S559 and the other between D10S1713 and D10S189. Most of 10q deletions were large and included a region distal to D10S554. Four glioblastomas of 122 had patterns suggestive of homozygous deletions at D10S541, a locus close and distally located to the PTEN/MMAC1 gene. Losses of alleles were found not only in glioblastomas (93%) but also in anaplastic astrocytomas (66%) and in astrocytomas (35%). Most glioblastomas lost one entire chromosome 10, while astrocytomas preferentially lost only 10p. The data suggest that a number of tumor suppressor genes on chromosome 10, in addition to PTEN/MMAC1, may be associated with astrocytic glioma development.

Chromosome 10 deletion mapping in human gliomas: a common deletion region in 10q25.

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The high incidence of loss of chromosome 10 alleles in glioblastoma multiforme suggests the presence on this chromosome of a tumor suppressor gene that is important in glioma tumorigenesis and progression. Our initial deletion mapping studies using restriction fragment length polymorphism markers indicated a common deletion region in 10q24-qter. In an attempt to localize the deleted region further, we screened a panel of 117 gliomas for loss of heterozygosity for chromosome 10 loci using 10 microsatellite markers. Seventeen tumors showed partial loss of a copy of chromosome 10 and were further analysed using 28 additional microsatellite markers. Of these, 10 had terminal deletion in the q arm, three had deletions in both p and q arms, two contained interstitial deletion in 10q and two carried deletions in 10p. In the 15 tumors with deletions in 10q, the minimal overlapping deletion region was in distal 10q between markers D10S587 and D10S216. Loci D10S587 and D10S216 are approximately mapped to a 5 cM region in 10q25.1.

Association of epidermal growth factor receptor gene amplification with loss of chromosome 10 in human glioblastoma multiforme.

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Although the loss of tumor suppressor genes and the activation of oncogenes have been established as two of the fundamental mechanisms of tumorigenesis in human cancer, little is known about the possible interactions between these two mechanisms. Loss of genetic material on chromosome 10 and amplification of the epidermal growth factor receptor (EGFR) gene are the most frequently reported genetic abnormalities in glioblastoma multiforme. In order to examine a possible correlation between these two genetic aberrations, the authors studied 106 gliomas (58 glioblastomas, 14 anaplastic astrocytomas, five astrocytomas, nine pilocytic astrocytomas, seven mixed gliomas, six oligodendrogliomas, two ependymomas, one subependymoma, one subependymal giant-cell astrocytoma, and three gangliogliomas) with Southern blot analysis for loss of heterozygosity on both arms of chromosome 10 and for amplification of the EGFR gene. Both the loss of genetic material on chromosome 10 and EGFR gene amplification were restricted to the glioblastomas. Of the 58 glioblastoma patients, 72% showed loss of chromosome 10 and 38% showed EGFR gene amplification.

The remaining 28% had neither loss of chromosome 10 nor EGFR gene amplification. Without exception, the glioblastomas that exhibited EGFR gene amplification had also lost genetic material on chromosome 10 (p less than 0.001). This invariable association suggests a relationship between the two genetic events. Moreover, the presence of 15 cases of glioblastoma with loss of chromosome 10 but without EGFR gene amplification may further imply that the loss of a tumor suppressor gene (or genes) on chromosome 10 precedes EGFR gene amplification in glioblastoma tumorigenesis.

Deletion mapping of gliomas suggest the presence of two small regions for candidate tumor-suppressor genes in a 17-cM interval on chromosome 10q.

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The loss of genetic material on chromosome 10q is frequent in different tumors and particularly in malignant gliomas. We analyzed 90 of these tumors and found loss of heterozygosity (LOH) in >90% of the informative loci in glioblastoma multiforme (GBM). Initial studies restricted the common LOH region to 10q24-qter. Subsequently, the study of a pediatric GBM suggested D10S221 and D10S209, respectively, as centromeric and telomeric markers of a 4-cM LOH region. It is interesting to note that, in one subset of cells from this tumor, locus D10S209 seems involved in the allelic imbalance of a larger region, with D10S214 as telomeric marker. This 17-cM region contains the D10S587-D10S216 interval of common deletion recently defined on another set of gliomas.

New deletion in low-grade oligodendroglioma at the glioblastoma suppressor locus on chromosome 10q25-26.

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Loss of heterozygosity on chromosome 10 is considered to be associated with the progression of glioblastomas. Two closely related regions have recently been proposed to contain the glioblastoma suppressor locus on chromosome 10q25-26; a 1 cM region between the polymorphic (CA)n-repeat markers D10S587 and D10S216, and an area of 5 cM between the markers D10S221 and D10S209. To confirm and further delineate this region, we analyzed 51 glioblastomas and 11 intermediate and low-grade gliomas for loss of heterozygosity on chromosome 10. 47/62 mostly malignant gliomas displayed complete loss of chromosome 10 and nine tumors were unaltered, whereas four glioblastomas and two low-grade oligodendrogliomas had partial loss on distal 10q. With these six tumors, we constructed a deletion map with increased marker density at 10q25-26 which shows two centromeric breakpoints at the markers D10S587 and D10S216, thus only confirming the distal, but not the proximal candidate glioblastoma suppressor locus. Two out of four low-grade oligodendrogliomas displayed partial deletions on 10q25-26. This suggests that deletion on chromosome 10 is not merely a late event in the progression of glioblastomas, but could play a role earlier in the development of gliomas.

p53 mutation and loss of heterozygosity on chromosomes 17 and 10 during human astrocytoma progression.

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The human brain tumor, astrocytoma, typically progresses through three histopathologically defined stages with the passage of time: one premalignant stage, low-grade astrocytoma; and two malignant stages, anaplastic astrocytoma and glioblastoma multiforme. We correlated the results of a sequence analysis of the tumor suppressor gene, p53, and a restriction fragment length polymorphism analysis of chromosomes 17 and 10 in 45 patients with cerebral astrocytomas at different stages. To detect p53 mutations in tumor DNA, we analyzed polymerase chain reaction products corresponding to every p53-coding exon for single-strand conformation polymorphisms and confirmed the mutations by sequencing. Loss of heterozygosity (LOH) was determined by Southern transfer analysis of somatic and tumor DNA from these same patients using polymorphic markers for various loci on chromosomes 10 and 17. p53 mutations were found in 7 of 25 glioblastomas (28%), in 5 of 14 anaplastic astrocytomas (36%) but in 0 of 6 low-grade astrocytomas. p53 mutations were found in 62% of patients with LOH on chromosome 17p. These results indicated that p53 inactivation is a common genetic event in astrocytoma progression that may signal the transition from benign to malignant tumor stages. LOH on chromosome 10 was found in 61% of glioblastomas, in 23% of anaplastic astrocytomas, but in 0% of low-grade astrocytomas. LOH on chromosome 10 and p53 mutation were found together only in patients with glioblastoma multiforme (22%), suggesting that these genetic changes may accumulate during astrocytoma progression.

Roles of the functional loss of p53 and other genes in astrocytoma tumorigenesis and progression.

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Loss of function of the p53 tumor suppressor gene due to mutation occurs early in astrocytoma tumorigenesis in about 30-40% of cases. This is believed to confer a growth advantage to the cells, allowing them to clonally expand due to loss of the p53-controlled G1 checkpoint and apoptosis. Genetic instability due to the impaired ability of p53 to mediate DNA damage repair further facilitates the acquisition of new genetic abnormalities, leading to malignant progression of an astrocytoma into anaplastic astrocytoma. This is reflected by a high rate of p53 mutation (60-70%) in anaplastic astrocytomas. The cell cycle control gets further compromised in astrocytoma by alterations in one of the G1/S transition control genes, either loss of the p16/CDKN2 or RB genes or amplification of the cyclin D gene. The final progression process leading to glioblastoma multiforme seems to need additional genetic abnormalities in the long arm of chromosome 10; one of which is deletion and/or functional loss of the PTEN/MMAC1 gene. Glioblastomas also occur as primary (de novo) lesions in patients of older age, without p53 gene loss but with amplification of the epidermal growth factor receptor (EGFR) genes. In contrast to the secondary glioblastomas that evolve from astrocytoma cells with p53 mutations in younger patients, primary glioblastomas seem to be resistant to radiation therapy and thus show a poorer prognosis. The evaluation and design of therapeutic modalities aimed at preventing malignant progression of astrocytomas and glioblastomas should now be based on stratifying patients with astrocytic tumors according to their genetic diagnosis.

Genetic pathways to primary and secondary glioblastoma.

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Glioblastoma is the most frequent and most malignant human brain tumor. The prognosis remains very poor, with most patients dying within 1 year after diagnosis. Primary and secondary glioblastoma constitute distinct disease subtypes, affecting patients of different age and developing through different genetic pathways. The majority of cases (>90%) are primary glioblastomas that develop rapidly de novo, without clinical or histological evidence of a less malignant precursor lesion. They affect mainly the elderly and are genetically characterized by loss of heterozygosity 10q (70% of cases), EGFR amplification (36%), p16(INK4a) deletion (31%), and PTEN mutations (25%). Secondary glioblastomas develop through progression from low-grade diffuse astrocytoma or anaplastic astrocytoma and manifest in younger patients. In the pathway to secondary glioblastoma, TP53 mutations are the most frequent and earliest detectable genetic alteration, already present in 60% of precursor low-grade astrocytomas. The mutation pattern is characterized by frequent G:C-->A:T mutations at CpG sites. During progression to glioblastoma, additional mutations accumulate, including loss of heterozygosity 10q25-qter (approximately 70%), which is the most frequent genetic alteration in both primary and secondary glioblastomas. Primary and secondary glioblastomas also differ significantly in their pattern of promoter methylation and in expression profiles at RNA and protein levels. This has significant implications, particularly for the development of novel, targeted therapies, as discussed in this review.